Huawei Access Node HEB-ACC-33 Hebbal Edge Hardware/Environmental Fault Resolution

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4183

Description

The access node, HEB-ACC-33, has encountered a hardware/environmental fault due to ASIC memory parity errors or line card crashes, resulting in a power supply redundancy loss. This failure is attributed to the specific role and vendor equipment as a result of the inherent architecture of the Huawei Access Node, where the ASIC memory parity errors can lead to a cascade of failures, ultimately resulting in a line card crash. Furthermore, the power supply redundancy loss is a critical failure that can occur due to various environmental factors such as power surges, overheating, or electrical noise. The underlying protocol or hardware mechanics of the Access Node's control and data planes are designed to ensure seamless communication and high availability, but in this instance, the hardware/environmental fault has compromised these capabilities, necessitating a thorough diagnostic and resolution process to rectify the issue.

Pre-requisite Checks

- Confirm the site location of the faulty node as Hebbal Edge

- Verify the Node ID as HEB-ACC-33

- Check the redundant link status to ensure it is operational

- Dispatch a remote hands team to the Hebbal Edge site to facilitate physical checks and interventions if necessary

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	High

Diagnostic Commands & Expected Outputs

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Command: display current-temperature, Expected Output: Current temperature of the node, Description: This command is used to verify the temperature of the node, which can be an indicator of overheating or thermal issues that may be contributing to the hardware/environmental fault.

Command: display power-supply, Expected Output: Power supply redundancy status, Description: This command is used to verify the power supply redundancy status, which can help identify if the power
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supply redundancy loss is the root cause of the failure.

Command: display diagnose-asic-memory-parity, Expected Output: ASIC memory parity error information, Description: This command is used to verify the ASIC memory parity error information, which can help identify if the ASIC memory parity errors are the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace line-card {card-name}, Description: This step involves replacing the faulty line card with a new one to rectify the line card crash issue. The {card-name} should be replaced with the actual name of the faulty line card.

Step 2: Step: Replace the faulty power supply, Command: replace power-supply {power-supply-name}, Description: This step involves replacing the faulty power supply with a new one to rectify the power supply redundancy loss issue. The {power-supply-name} should be replaced with the actual name of the faulty power supply.

Step 3: Step: Perform a firmware upgrade, Command: upgrade firmware {firmware-version}, Description: This step involves performing a firmware upgrade to the latest version to rectify any potential firmware-related issues. The {firmware-version} should be replaced with the actual version number of the latest firmware.

Post-Check Verification

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Command: display current-temperature, Expected Output: Current temperature of the node within normal operating range, Description: This command is used to verify that the temperature of the node is within the normal operating range after the resolution steps have been performed.

Command: display power-supply, Expected Output: Power supply redundancy status indicating redundancy is operational, Description: This command is used to verify that the power supply redundancy status indicates that the redundancy is operational after the resolution steps have been performed.

Command: display diagnose-asic-memory-parity, Expected Output: ASIC memory parity error information indicating no errors, Description: This command is used to verify that the ASIC memory parity error information indicates that there are no errors after the resolution steps have been performed.
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Roll-back Procedures

- Step: Restore the original configuration, Command: restore config {config-file}, Description: This step involves restoring the original configuration from the {config-file} to revert any changes made during the resolution process.

- Step: Replace the replacement component with the original component, Command: replace {component-name} with {original-component-name}, Description: This step involves replacing the replacement component with the original component to revert any changes made during the resolution process. The {component-name} and {original-component-name} should be replaced with the actual names of the components.